Assignments 4 & 5, 2026

Level: BE

Subject : Applied Mathematics

Due Date : July 20, 2026

Attempt all questions.

1. Classify the following PDEs.

   a. $3\frac{\partial^2 u}{\partial x^2} + 6\frac{\partial^2 u}{\partial x \partial y} + 3\frac{\partial^2 u}{\partial y^2} = 0$

   b. $\frac{\partial^2 u}{\partial x^2} + x\frac{\partial^2 u}{\partial t \partial x} + t\frac{\partial^2 u}{\partial t^2} + 2\frac{\partial u}{\partial t} + 2u = 0$

2. Solve the PDE

   a. $u_y + 2yu = 0$

   b. $u_{yy} = u_y$

   c. $u_{xy} = u_x$

3. Find $u(x,t)$ of the string of length $L = \pi$ when $c^2 = 1$, the initial velocity is zero and the initial deflection is $k(sinx - \frac{1}{2} sin\, 2x)$.

4. A tightly stretched string of length L is drawn a side at its midpoint a distance $L/2$ perpendicular to the equilibrium position s that its initial position is given by

$$f(x) \begin{cases} x & \text{if } 0 < x < L/2 \\ (L-x) & \text{if } L/2 < x < L \end{cases}.$$

   It is initially at rest and suddenly released. Find $u(x,t)$.

5. A tight string of length 20cm fastened at both ends is displaced from its position of equilibrium by imparting to each of its points at initial velocity

$$g(x) = \begin{cases} x & \text{if } 0 \le x \le 10 \\ 20-x & \text{if } 10 \le x \le 20 \end{cases}.$$

   Find the displacement at any time t. (Hint: $B_n = 0$)

6. Solve the following PDEs by separation of variables method.

   a. $3u_x + 2u_y = 0$

   b. $xu_x - 4yu_y = 0$

   c. $y^2 u_x - x^2 u_y = 0$

   d. $u_x + u_y = (x+y)u$

7. Find the temperature $u(x,t)$ in a bar of silver bar [length 10cm, constant cross section area $1cm^2$, density $10.6gm/cm^3$, thermal conductivity $1.04cal/(gm\ sec^0 c)$, specific heat $0.056\, cal/(gm^0 c)$] that is perfectly insulated laterally, whose ends are kept at temperature $0^0 c$ and whose initial temperature $(in^0 c)$ is $f(x)$, where

   a. $f(x) = sin(0.1)\pi x$ b. $f(x) = x(10-x)$. [Hint: $c^2 = \frac{K}{s\rho}$]

8. Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature 0, assuming that the initial temperature is

$$f(x) = \begin{cases} x & \text{if } 0 < x < L/2 \\ L - x & \text{if } L/2 < x < L \end{cases}.$$

9. Solve the one dimensional heat equation $u_t = c^2 u_{xx}$ under the condition
   i. u is finite at $t \to \infty$
   ii. $u(0,t) = 0 = u(\pi, t)$
   iii. $u(x, 0) = \pi x - x^2$

10. A rod of length L has its ends A and B kept at 0^0c and 100^0c respectively until steady state condition prevails. If the change consists of raising the temperature of A to 25^0c and reducing that of B to 75^0c. Find $u(x,t)$.

11. Solve the PDE $u_{xx} + u_{yy} = 0$, $0 < x < \pi$, $0 < y < \pi$ which satisfies the conditions $u(0, y) = u(\pi, y) = u(x, \pi) = 0$ and $u(x, 0) = sin^2 x$.

12. A rectangular plate with insulated surface is 8cm wide and so long compared to its width that it may be considered infinite in length without introducing an appreciable. If the temperature along one short edge $y = 0$ is given by $u(x, 0) = 100 sin(\frac{\pi\, x}{8})$, $0 < x < 8$
while the two long edges $x = 0$ and $x = 8$ as well as the other short edge are kept at 0^0c, show that the steady state temperature at any point of plate is given by $u(x, y) = 100 e^{-\frac{\pi\, y}{8}} sin(\frac{\pi\, x}{8})$.

13. Show that

$$\int_0^\infty [\frac{sin\, w}{2w} cos\, wx + (\frac{1 - cos\, w}{2w}) sin\, wx] dw = \begin{cases} \pi/2 & \text{if } 0 \le x < 1 \\ \pi/4 & \text{if } x = 1 \\ 0 & \text{if } x > 1 \;\&\; x < 0 \end{cases}.$$

14. Show that

$$\int_0^\infty [\frac{cos\, \frac{\pi}{2}\, cos\, wx}{1 - w^2}] dw = \begin{cases} \frac{\pi}{2} cos\, x & \text{if } |x| < \pi/2 \\ 0 & \text{if } |x| > \pi/2 \end{cases}.$$

15. Show that

$$\int_0^\infty [\frac{sin\, \pi\, w\;\; sin\, wx}{1 - w^2}] dw = \begin{cases} \frac{\pi}{2} sin\, x & \text{if } 0 \le x \le \pi \\ 0 & \text{if } x > \pi \end{cases}.$$

16. Show that

$$\int_0^\infty [\frac{sin\, \pi\, w\;\; sin\, wx}{w}] dw = \begin{cases} \frac{\pi}{2} sin\, \pi\, x & \text{if } 0 \le x \le \pi \\ 0 & \text{if } x > \pi \end{cases}.$$

17. Find the Fourier Sine transform of $f(x) = (\frac{x^3}{x^4 + 4})$.

18. Find Fourier cosine and sine integral representations of $f(x) = e^{-kx}$ for $x > 0$, $k > 0$

19. Find the Fourier cosine integral representation of

$$f(x) = \begin{cases} x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}.$$

20. Find the Fourier sine transform of $f(x) = e^{-|x|}$ and hence evaluate $\int_0^\infty \frac{w\, sin\, wx}{1+w^2}\, dw$.

21. Find the Fourier sine and cosine transforms of $f(x) = e^{-x},\ x > 0$. Also, show that

$$\int_0^\infty \frac{x\ sin\, mx}{x^2+1}\, dx = \frac{\pi}{2} e^{-m},\ m > 0.$$

22. Write the sufficient condition for the existence of Fourier transform of a function $f(x)$. Find the Fourier transform of

$$f(x) = \begin{cases} |x| & \text{if } -1 < x < 1 \\ 0 & \text{if } otherise \end{cases}.$$

23. Find the Fourier transform of $f(x) = e^{-x^2}$.

24. Find the Fourier transform of $f(x) = x\, e^{-x^2}$.

25. Verify the convolution theorem for the functions $f(x) = g(x) = e^{-x^2}$

26. Find the Fourier sine and cosine transforms of

$$f(x) = \begin{cases} 1 & \text{if } 0 \le x \le 1 \\ 0 & \text{if } x > 1 \end{cases}.$$

and then by using Parseval's Identities show that
i. $\int_0^\infty \left(\frac{1-cos\, x}{x}\right)\, dx = \frac{\pi}{2}$ ii. $\int_0^\infty \left(\frac{sin\, x}{x}\right)^2\, dx = \frac{\pi}{2}$

27. Find the Fourier Sine transform of $f(x) = \frac{x}{x^2+1}$, then using Parseval's identity show that

$$\int_0^\infty \frac{x^2}{(x^2+1)^2} dx = \frac{\pi}{4}.$$

(Hint: Use question number 21)

28. Find the Fourier transform of

$$f(x) = \begin{cases} 1 & \text{if } |x| < a \\ 0 & \text{if } |x| > a \end{cases}.$$

and then by using Parseval's Identity show that
$\int_{-\infty}^\infty \left(\frac{sin^2\, wx}{w^2}\right)\, dw = \pi\, a.$

29. Find the Fourier transform of

$$f(x) = \begin{cases} 1 - |x| & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}.$$

and then by using Parseval's Identity show that

$$\int_0^\infty \left(\frac{sin^4\, t}{t^4}\right)\, dt = 4\pi\,/3.$$

(Hint: put $\frac{w}{2} = t$)

30. Solve the integral equation

$$\int_0^\infty f(t)\ cos\, wt\ dt = \begin{cases} 1 - w & \text{if } 0 \le w \le 1 \\ 0 & \text{if } w > 1 \end{cases}.$$

"ALL THE BEST."